

Problem

The voltage V_o across the capacitor in Fig. 10.43 is:

- (a) $5\angle 0^\circ \text{ V}$ (b) $7.071\angle 45^\circ \text{ V}$
 (c) $7.071\angle -45^\circ \text{ V}$ (d) $5\angle -45^\circ \text{ V}$

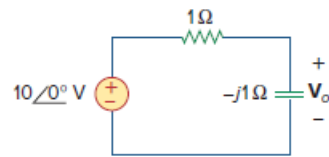


Figure 10.43
For Review Question 10.1.

Step-by-step solution

Step 1 of 2

Refer to Figure 10.43 for the circuit in the textbook.

Redraw the circuit as shown in Figure 1.

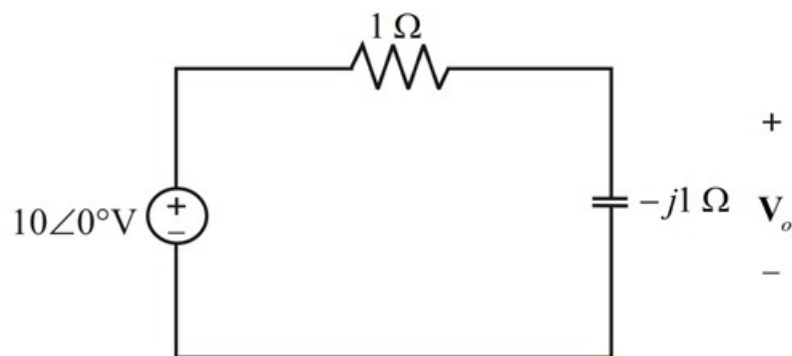


Figure 1

Step 2 of 2

Apply the voltage division rule to the circuit and obtain the voltage across the capacitor.

$$\begin{aligned} \mathbf{V}_o &= (10\angle 0^\circ) \left(\frac{-j}{1-j} \right) \\ &= (10\angle 0^\circ) \left(\frac{\sqrt{0^2 + (-1)^2} \angle \tan^{-1} \left(\frac{-1}{0} \right)}{\sqrt{1^2 + (-1)^2} \angle \tan^{-1} \left(\frac{-1}{1} \right)} \right) \\ &= (10\angle 0^\circ) \left(\frac{\sqrt{0^2 + (-1)^2} \angle -\tan^{-1}(\infty)}{\sqrt{1^2 + (-1)^2} \angle -\tan^{-1}(1)} \right) \\ &= (10\angle 0^\circ) \left(\frac{1\angle -90^\circ}{1.414\angle -45^\circ} \right) \\ &= (10\angle 0^\circ)(0.7071\angle -45^\circ) \\ &= 7.071\angle -45^\circ \text{ V} \end{aligned}$$

Therefore, the voltage across the capacitor $\mathbf{V}_o$ is $7.071\angle -45^\circ \text{ V}$.

Therefore, the correct option is (c).